

Name _____ Surname _____ Date ____/____/____

Midterm test, Term 1 — Speed, energy and work

Version A

Time: 75 minutes. Calculator and formula sheet (last page) allowed. Use $g = 10 \text{ N/kg}$. Write on separate sheets, show every step of your working and write the unit in each answer. Give answers to 3 significant figures where needed. Total: 55 marks.

 **Lifebuoy.** The parts marked with the lifebuoy are worth 33 marks out of 55, that is 60%: if you have completed all of them, you are safe. The other parts take you beyond the pass mark.

Exercise 1 — Multiple choice

 Only one answer is correct. Each question is worth 1 mark, 8 marks in total.

1. The energy stored in a battery is:
A kinetic B chemical C elastic (strain) D gravitational potential
2. A train travels 240 km in 3.0 h. Its average speed is:
A 0.0125 km/h B 720 km/h C 80 km/h D 243 km/h
3. On a distance–time graph, a steeper line means that the object is moving:
A more slowly B backwards C at rest D faster
4. A ball is thrown upwards. At the top of its path, ignoring air resistance, the ball has:
A maximum kinetic energy B maximum gravitational potential energy C zero total energy
D maximum speed
5. The mass of a trolley is doubled and its speed stays the same. Its kinetic energy:
A doubles B stays the same C halves D becomes four times bigger
6. In which situation is work being done on the object?
A a table supports a book B a boy holds a bag still C a dog pulls a sledge along the snow
D a wall holds up a roof
7. A force of 5.0 N moves an object 3.0 m in the direction of the force. The work done is:
A 1.7 J B 8.0 J C 15 N D 15 J
8. A cyclist has 500 J of kinetic energy. After braking, 120 J of it has been transferred to thermal energy. The kinetic energy left is:
A 380 J B 620 J C 120 J D 500 J

Exercise 2 — True or false

 Write T or F. No explanation needed. Each statement is worth 1 mark, 4 marks in total.

- (a) If the speed of an object doubles, its kinetic energy doubles.
- (b) Energy can be stored and transferred, but its total amount stays the same.
- (c) A girl pushes a wall that does not move: she does work on the wall.
- (d) The unit of work is the joule, and $1 \text{ J} = 1 \text{ N m}$.

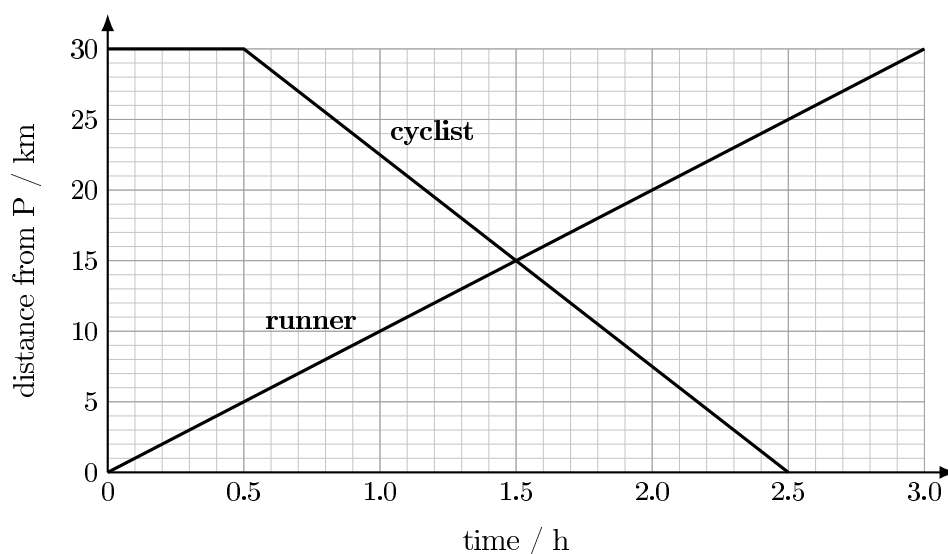
Exercise 3 — Energy stores

⚽ Complete the table. In each event, name the energy store that decreases and the one that increases. 1 mark for each row, 3 marks in total.

Event	Store that decreases	Store that increases
A girl releases a stretched bowstring and the arrow flies off.		
A cyclist freewheels down a hill without pedalling (ignore thermal energy).		
A student runs up a flight of stairs.		

Exercise 4 — Two travellers

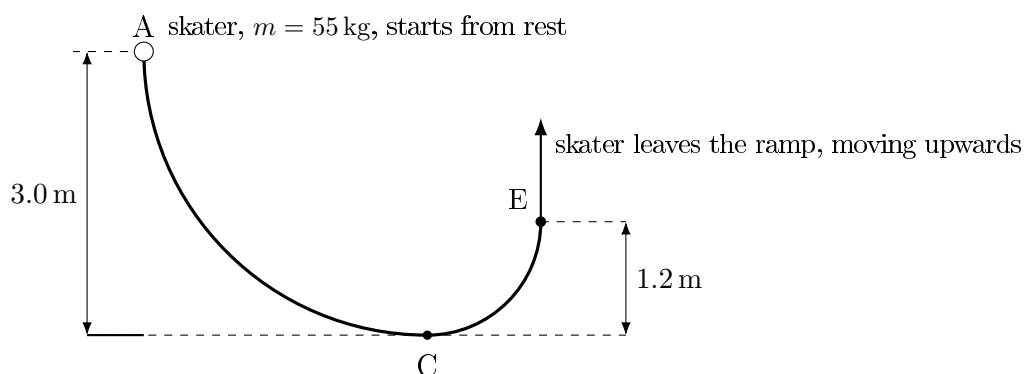
A runner and a cyclist travel on the same straight road between town P and town Q. The graph shows the distance of each of them from P. The runner starts from P and the cyclist starts from Q at the same moment.



- ⚽ Read the graph: for how long does the cyclist stay at rest at the start, and at what time and at what distance from P do the two meet? **[2]**
- Calculate the speed of the cyclist while she is moving, in km/h. **[2]**
- Calculate the average speed of the cyclist for the whole journey from Q to P, in km/h. **[2]**
- Calculate the speed of the runner in m/s. **[2]**

Exercise 5 — The half-pipe

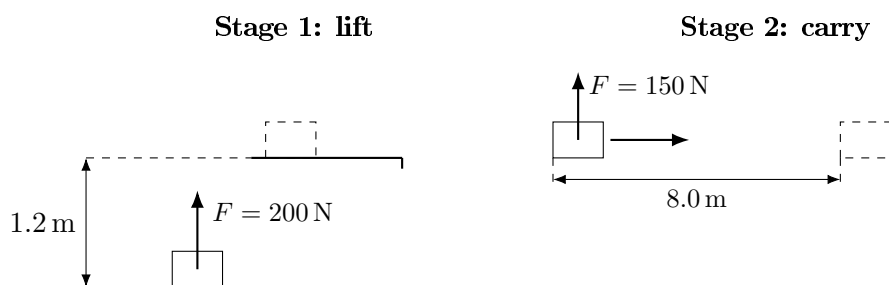
A skater of mass 55 kg starts from rest at A, at the top of a half-pipe, 3.0 m above the lowest point C. She rolls down, passes C and goes up the right-hand side, leaving the ramp at E, 1.2 m above C, moving vertically upwards. Ignore friction and air resistance.



- Without calculating, say how high above C the skater rises before she starts to fall, and justify your answer in one line. [2]
- Write, in symbols and without numbers, the equation that gives her speed v at E in terms of the heights h_A and h_E above C. [2]
- Calculate the gravitational potential energy lost between A and E. [2]
- Calculate her speed at E. [2]

Exercise 6 — Lifting and carrying a box

A removal worker lifts a box of mass 15 kg from the floor to a shelf 1.2 m above it (stage 1). His hands push upwards on the box with a constant force of 200 N, and the box is still moving upwards when it reaches the height of the shelf. In stage 2 he walks 8.0 m along a corridor, holding the box at that height with a force of 150 N from his hands, pointing upwards.



- Write, in symbols, the equation for the work done by his hands and the equation for the gain in gravitational potential energy of the box in stage 1. [2]
- Calculate the work done by his hands on the box and the gain in gravitational potential energy of the box in stage 1. [2]
- The work done is greater than the gain in gravitational potential energy. In which energy store does the difference end up? [1]
- State the work done by his hands on the box in stage 2. [1]
- Explain your answer to (d) in one or two lines. (His muscles use chemical energy even while he carries the box: here we ask only about the work done on the box.) [2]

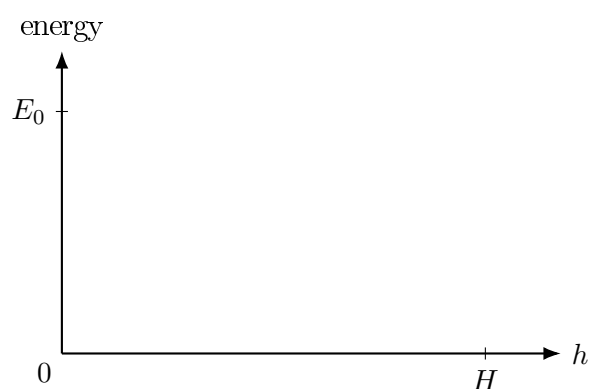
Exercise 7 — A car brakes

A car of mass 1200 kg travels at 15 m/s. The driver brakes and the car stops after 25 m. Assume that the braking force is constant.

- (a) ☐ In which energy store does the kinetic energy of the car mainly end up? [1]
- (b) Calculate the kinetic energy of the car before braking. [2]
- (c) Calculate the braking force. [2]
- (d) ☐ The car now carries passengers, so that its mass is doubled. It travels at the same speed and the braking force is the same. The stopping distance becomes: the same, twice as long, four times as long, eight times as long. Circle the correct answer and give a reason in one line. [1]

Exercise 8 — A stone thrown upwards

A stone of mass 0.20 kg is thrown vertically upwards from the ground at 12 m/s. Ignore air resistance. On the axes below, E_0 is the kinetic energy at launch and H is the maximum height.



- (a) ☐ Draw and label how the kinetic energy and the gravitational potential energy change with the height h , from $h = 0$ to $h = H$. [2]
- (b) ☐ Write, in symbols and without numbers, the equation that gives the maximum height H . [2]
- (c) Calculate the kinetic energy at launch. [2]
- (d) Calculate the maximum height. [2]
- (e) ☐ The same object is thrown with twice the speed. The maximum height becomes: the same, twice as big, four times as big, eight times as big. Circle the correct answer and give a reason in one line. [2]